QP CODE:25901657

MGU-UGP(HONOURS)REGULAR EXAMINATION,APRIL 2026

SECOND SEMESTER

DISCIPLINE SPECIFIC CORE COURSE(DSC)-MG2DSCECT101-DATA COMMUNICATION

(2024 ADMISSION ONWARDS)

Remember(K),Understand(U), Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S) and Appreciation(AP)

SCHEME OF EVALUATION

PART A

Multiple choice Questions

Answer all questions.

Each Carries 1 marks

		Learning Domain	CO
1.	a.Amplitude distortion	[U]	[CO1]
2.	c.Digital Signals are less affected by noise and can be processed by computers	[K]	[CO1]
3.	a.The number of bits transmitted per unit time	[U]	[CO1]
4.	d.Plastic or foam	[k]	[CO2]
5.	a.Signals cannot pass through solid objects	[U]	[CO2]
6.	b.Simpler wiring and less signal interference	[U]	[CO3]
7.	b.Frequency of the carrier signal	[U]	[CO3]
8.	b.Introduce redundancy for error detection and correction	[U]	[C03]
9.	b.Twice base bandwidth	[U]	[CO3]
10.	c.Baud rate represents the number of signal units per second,while bit rate represents the number of bits per second	[U]	[CO3]

PART B

Short Answer Questions

Answer any four questions

Each Questions carries 5 Marks

11. **Periodic and Non-periodic signals** [U] [CO1]

Definition of periodic signal with example– 2 mark
Definition of non-periodic signal with example – 2 mark
Diagram of periodic&non-periodic signals – 1 mark

12. Types of Data Flow **[U] [C01]**

Definition of Simplex – 1 mark
Definition of Half-Duplex – 1 mark
Definition of Full-Duplex – 1 mark
Diagram – 2 mark

13. Applications of Fiber Optic Communication **[K] [CO2]**

Any five valid applications (1 mark each):
Examples:
Internet communication
Cable TV
Medical endoscopy
Military communication
Telephone networks
Data centers

14. Compare FDM and WDM **[U] [CO3]**

Definition of FDM – 1 mark
Definition of WDM – 1 mark
Key differences (any 3 points × 1 mark each):
Medium used
Signal type
Application area
Frequency vs wavelength

15. Benefits of Spread Spectrum Technology **[K] [CO3]**

Any five points (1 mark each):
Resistance to interference
Security (hard to intercept)
Anti-jamming capability
Multiple access support
Noise immunity
Reliable communication

16. Data Framing: Synchronous vs Asynchronous **[U] [CO3]**

Definition of synchronous transmission – 1 mark
Definition of asynchronous transmission – 1 mark
Framing in synchronous – 1 mark
Framing in asynchronous – 1 mark
Key difference (timing/efficiency/overhead) – 1 mark

PART C

Essay Questions

Answer any 2 Questions

Each Questions carries 10 Marks.

17. Data Rate Limits & Formulas [U] [CO1]

- **Explanation of Bandwidth, Signal Levels, and Noise -2 Marks**
- **Nyquist Bitrate =2*B*log2(L) and application to noiseless channels- 3 Marks**
- **Shannon Capacity C=B*log2(1+SNR)and application to noisy channel-3 Marks**
- **Diagram- 2 Marks**

18. Twisted Pair Cable Construction [U] [CO2]

- **Construction Description -2 Marks**
- **Technical Diagram (A labeled sketch showing the internal wires,the twist,and the protective layers) -4Marks**
- **UTP vs STP Comparison- 4 Marks**

19. Direct Sequence Spread Spectrum (DSSS) [U] [CO3]

- **Definition: Definition of spreading the signal over a wider bandwidth – 3 Marks**
- **Block Diagram: A Functional diagram showing the Data Input,PN Code Generator,and the Multiplier /Modulator – 4 Marks**
- **Bandwidth Comparison: Showing the narrowband vs "spread" signal in the frequency domain – 3 Marks**